Software Requirements Specification

for

Rihla

Version 1.0 approved

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Revision History

Name	Date	Reason For Changes	Version
Yara Almozainy	April 2026	Initial Release	1.0

1. Introduction

1.1 Purpose

This document, the Software Requirements Specification (SRS), specifies the functional and non-functional requirements for Rihla, a smart, AI-based tourism platform for Saudi Arabia. This SRS is for the software development team, project stakeholders and the supervisor at Prince Sultan University. It provides a description of what the system shall do and the constraints imposed on it.

1.2 Document Conventions

- “Shall” is a mandatory requirement.
- “Should" indicates a recommended but not mandatory requirement.
- The priority of the requirement is High, Medium or Low.
- Subrequirements have the same priority as their parent unless otherwise indicated.

1.3 Intended Audience and Reading Suggestions

This document is intended for:

- Developers - to gain insights into the functional requirements and system constraints.
- Project Supervisors/Professors - to assess the breadth and depth of the system.
- Testers - to develop test cases.
- Stakeholders - to ensure business objectives are achieved.

Developers and testers can concentrate on Sections 3 and 4. Stakeholders can focus on Sections 1 and 2.

1.4 Product Scope

<Provide a short description of the software being specified and its purpose, including relevant benefits, objectives, and goals. Relate the software to corporate goals or business strategies. If a separate vision and scope document is available, refer to it rather than duplicating its contents here.>

1.5 References

Rihla is an intelligent tourism platform for domestic and international tourists visiting Saudi Arabia. It offers a seamless digital platform that combines real-time transportation, personalised itinerary planning, smart recommendations, pricing and language support. Rihla tackles the siloed nature of existing tourism services by providing all the necessary travel services in one platform, in line with Saudi Vision 2030's aim to expand the tourism industry.

2. Overall Description

2.1 Product Perspective

Rihla is a mobile app designed to be a new standalone system that is part of the digital tourism ecosystem in Saudi Arabia. It communicates with third-party APIs for mapping, transportation, payments and tourism information. The system does not replace any existing system but adds value to the market by integrating various tourism services.

2.2 Product Functions

The key features of Rihla are:

- Personalised itinerary planning using AI algorithms.
- Real-time transport monitoring and information (bus, taxis, ride-shares).
- Location and event recommendations with extensive details and multimedia.
- Price estimation and information for travel and events.
- Multiple languages (Arabic and English).
- Ratings and reviews and recommendations from users.
- In-app booking and payment.

2.3 User Classes and Characteristics

- International Tourists: Users who are not familiar with Saudi Arabia and require translation, cultural insights and itinerary recommendations. Potentially low service familiarity.

- Domestic Tourists: Saudis or expats visiting local places. Moderate to high familiarity with local services; exploring and convenience seeking.
- Agencies/Travel Planners: Professional users who may plan events and trips for others. High familiarity with travel tools. classes. Distinguish the most important user classes for this product from those who are less important to satisfy.>

2.4 Operating Environment

- Mobile devices: iOS/ Android phones.
- Server: Cloud server.
- Network Access: Needed for real-time data; some offline functionality for itineraries.
- Third-party Services: Maps APIs (e.g., Google Maps), payment processing and transportation services.

2.5 Design and Implementation Constraints

- Data privacy in line with Saudi Arabia's regulations.
- The availability of third party APIs may impact real-time travel and booking.
- The system should support Arabic (RTL) and English (LTR) UI/UX.
- The system should be scalable to support a large number of tourists during peak periods (e.g., Hajj, national holidays).

2.6 User Documentation

The following documentation shall be provided:

- App Onboarding for new users.
- Frequently asked questions and Help page in the app.
- Online user manual link on the app’s help page.

2.7 Assumptions and Dependencies

- The user has a smartphone with internet connectivity.
- The system relies on external APIs (maps, transport, payments); any outages in these services could impact the system.
- It is assumed that current tourism data is available via public or commercial sources.
- Compliance with payment processing (e.g., PCI-DSS) is the responsibility of the payment gateway.

3. External Interface Requirements

3.1 User Interfaces

3.1 AI-Powered Itinerary Generation
Description and Priority: High priority. This is the main thing Rihla does. The system will create custom travel plans according to user preferences, duration, budget and interests.
Stimulus/Response Sequences:

- The user enters travel preference (dates, interests, budget).
- The computer processes the information with an AI engine.
- The system outputs a day-to-day itinerary with recommended visits, transport and estimated expenses.
- Users can generate a new itinerary, modify and accept an itinerary.

Functional Requirements:

- REQ-1.1: The system shall enable users to specify the duration of the trip, the price range and the type of interests (culture, adventure, food, etc.).
- REQ-1.2: The system shall provide a full itinerary with places to visit, approximate travel time and expenses.
- REQ-1.3: The system shall provide an interface to edit itineraries.
- REQ-1.4: The system shall store itineraries in the user's account.

3.2 Hardware Interfaces

Description and Priority: High priority. The system shall offer users real-time transportation information to reach their destinations.
Stimulus/Response Sequences:

- The user chooses a destination or reviews his or her itinerary.
- Real-time transportation options are retrieved.
- The system shows routes, predicted times and fares.

Functional Requirements:

- REQ-2.1: The system shall show transportation options (bus, taxi, carpool) from one location to another.
- REQ-2.2: The system shall display estimated travel and arrival times.
- REQ-2.3: The system shall use at least one map/navigation API.
- REQ-2.4: The system shall provide user notification of transport changes or delays (if any).

3.3 Software Interfaces

Description and Priority: High priority. Users must be able to search, browse and filter tourist attractions, activities and events in Saudi Arabia.
Stimulus/Response Sequences:

- The user clicks on the discovery section or enters a destination.
- Results are displayed according to location, category and rating.
- The user clicks on an attraction for more details.

Functional Requirements:

- REQ-3.1: The system shall show a searchable and filterable list of attractions and events.
- REQ-3.2: Attractions shall have descriptions, images, location, hours and ratings.
- REQ-3.3: Filters shall be available for category, price, and location.
- REQ-3.4: The system shall display featured or popular attractions on the homepage.

3.4 Communications Interfaces

Description and Priority: Medium priority. The system shall offer cost estimates for trips, activities and transportation.
Functional Requirements:

- REQ-4.1: The system shall show estimated costs for each itinerary item.
- REQ-4.2: The system shall display a summary of the total cost.
- REQ-4.3: If the user changes the itinerary, the system shall dynamically update the cost estimates.

3.5 Multilingual Support

Description and Priority: High priority. The system shall be able to support Arabic and English language throughout the user interface.
Functional Requirements:

- REQ-5.1: Users shall be able to choose their language during the setup process.
- REQ-5.2: The system shall display all UI components, content and notifications in Arabic and English.
- REQ-5.3: The system shall have RTL layout for Arabic and LTR layout for English.

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3.6 Reviews and Ratings

Description and Priority: Medium priority. The system shall allow users to leave and view reviews and ratings for attractions and services.
Functional Requirements:

- REQ-6.1: The system shall permit logged in users to write a review and rating (1-5 stars) for any attraction.
- REQ-6.2: The system shall show the average rating and recent reviews for an attraction on its page.

- REQ-6.3: The system shall provide a means of reporting inappropriate reviews.

3.7 Booking and payment

Description and Priority: Medium priority. Users shall be able to book and pay for activities in the app.
Functional Requirements:

- REQ-7.1: Users shall be able to book attractions or tours available in the app.
- REQ-7.2: The system shall connect to a secure payment system.
- REQ-7.3: The system shall email or notify the user's person with a booking confirmation.
- REQ-7.4: The system shall provide users with a way to track their booking history.

4. External Interface Requirements

4.1 User Interfaces

- The app shall support both Arabic (RTL) and English (LTR) layouts, switchable from settings.

- The home screen shall feature a search bar, personalized recommendations, and quick-access categories such as historical sites, nature, and religious tourism.

- The itinerary view shall present a clear day-by-day layout with editable entries.

- The app shall follow accessibility guidelines (sufficient contrast, readable font sizes).

- Navigation shall use a bottom tab bar for primary sections (Home, Discover, My Trips, Profile).

- Error messages shall be displayed in the user’s selected language with a suggested corrective action.

4.2 Hardware Interfaces

- The system shall access the device's GPS/location services to support location-based features such as nearby attractions across Saudi regions (Riyadh, Jeddah, AlUla, NEOM, etc.).

- The system shall function on standard iOS and Android touch screen devices.

- No special or custom hardware is required by the system.

4.3 Software Interfaces

External System	Purpose	Data Exchange
Google Maps API	Navigation, route display, and location search across Saudi Arabia's regions including Riyadh, Jeddah, AlUla, and NEOM.	GPS coordinates, route details, estimated travel times, and place information.
Saudi Public Transport Authority (PTA) API	Provides real-time public transportation data including bus schedules and routes within Saudi cities.	Bus routes, real-time arrival times, and schedule updates.
Uber / Careem API	Ride-hailing integration, as Careem is the dominant ride-sharing platform in Saudi Arabia.	Fare estimates, driver availability, and booking confirmation status.
PayTabs / STC Pay / Mada	Secure in-app payment processing using locally supported Saudi payment methods familiar to both local and foreign tourists.	Transaction amounts, payment authorization status, and booking confirmation.
Firebase Authentication	Manages user registration, login, and session handling securely.	User credentials, authentication tokens, and session data.
Saudi Tourism Authority	Supplies verified and	Attraction names,

(STA) Content API	up-to-date tourist attraction and event information across Saudi regions, ensuring content aligns with official tourism guidelines.	descriptions, images, operating hours, location data, and pricing.
Firebase Cloud Messaging (FCM)	Delivers push notifications to users in both Arabic and English for booking confirmations, itinerary reminders, and transp ortation updates.	Notification content, delivery status, and device tokens.
Absher API	Identity verification for Saudi and non-Saudi users where applicable, supporting compliance with Saudi regulations.	User identity verification status and visa information.

4.4 Communications Interfaces

- All communication between the app and backend servers shall use HTTPS.
- All data in transit shall be encrypted using TLS 1.2 or higher, in compliance with Saudi Arabia's PDPL and CITC regulations.
- Push notifications shall be delivered via Firebase Cloud Messaging (FCM) in both Arabic and English.
- The app shall handle network timeouts gracefully and display an appropriate message to the user if the connection fails.
- The system shall use REST API communication with all external services.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- The home screen shall be displayed within 3 seconds on a typical 4G network.
- Itinerary load times shall take no longer than 5 seconds.
- The system shall be able to handle 10,000 users simultaneously.

- The update of real-time transportation information shall be every 60 seconds or less.

5.2 Safety Requirements

- The system shall not save payment information on the device.
- In the case of a payment failure, the system shall provide a warning to the user and not proceed with the payment.

5.3 Security Requirements

- User login shall be required to use the system's personalised services (email/password or social login).
- The system shall store user passwords in a secure format (e.g., bcrypt).
- The system shall have a 30-minute session timeout.
- The system shall adhere to Saudi privacy laws.
- User data shall not be disclosed to third parties without their consent.

5.4 Software Quality Attributes

- Usability: The application shall be used by a new user with no training and include a tour.
- Reliability: The system shall have 99.5% availability for key functions.
- Maintainability: Code shall adhere to documented coding guidelines for ease of maintenance.
- Scalability: The backend shall be horizontally scalable.
- Portability: The app shall support iOS (14+) and Android (10+) devices.

5.5 Business Rules

- BR-1: Only registered and authenticated users shall be able to generate, save, or modify itineraries.
- BR-2: Only registered users shall be able to submit reviews or ratings for attractions.
- BR-3: A user must agree to the terms and conditions upon registration before accessing any platform features, in compliance with Saudi Arabia’s Personal Data Protection Law (PDPL).
- BR-4: Booking cancellations must be made at least 24 hours before the scheduled activity to be eligible for a refund.
- BR-5: All prices shall be displayed in Saudi Riyals (SAR) as the default currency, with optional conversion for foreign tourists.

- BR-6: User-generated content (reviews, ratings) is subject to moderation in compliance with Saudi Arabia's anti-cybercrime regulations; reported content shall be reviewed within 48 hours.
- BR-7: The system shall not complete a booking if payment authorization fails.
- BR-8: Attraction and event data must be verified by an admin before being published, ensuring alignment with the Saudi Tourism Authority's (STA) approved destinations and guidelines.
- BR-9: Guest users may browse attractions but cannot save itineraries, make bookings, or process payments.
- BR-10: Transportation data displayed must not be older than 60 seconds to ensure accuracy.
- BR-11: The platform shall not promote or display content that conflicts with Saudi cultural values or Islamic guidelines.
- BR-12: During peak seasons such as Ramadan, Hajj, and Saudi National Day, the system shall prioritize high-availability mode to handle increased traffic.

6. Other Requirements

The following requirements are noted but details will be determined in later phases of development:

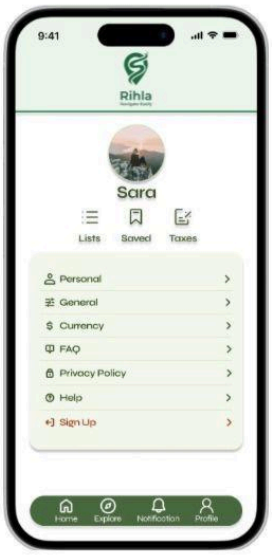
- Accessibility (WCAG 2.1 AA) for people with disabilities.
- Analytics and reporting dashboard for administrators to view usage and the most visited destinations.
- Offline access to view itineraries when offline.

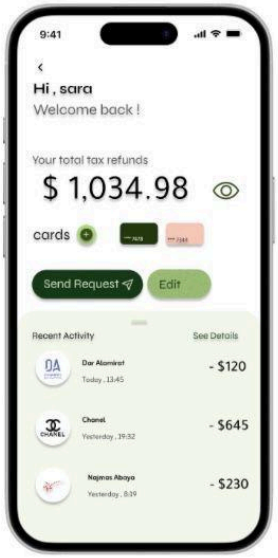
Appendix A: Glossary

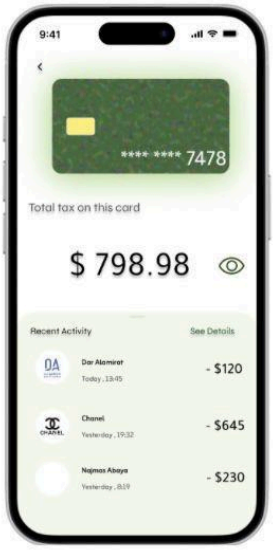
Term	Definition
Rihla	Arabic word for "journey"; the name of the tourism platform
AI	Artificial Intelligence; used for recommendations and trip planning
API	Application Programming Interface; used to integrate the app with other services

RTL	Right-to-Left; text direction used for the Arabic language
LTR	Left-to-Right; text direction used for the English language
SRS	Software Requirements Specification
Vision 2030	Saudi Arabia’s national development plan, including goals to grow the tourism sector
Itinerary	A planned route or schedule of travel activities

Appendix B: Analysis Models







Rihla
Navigate Easily

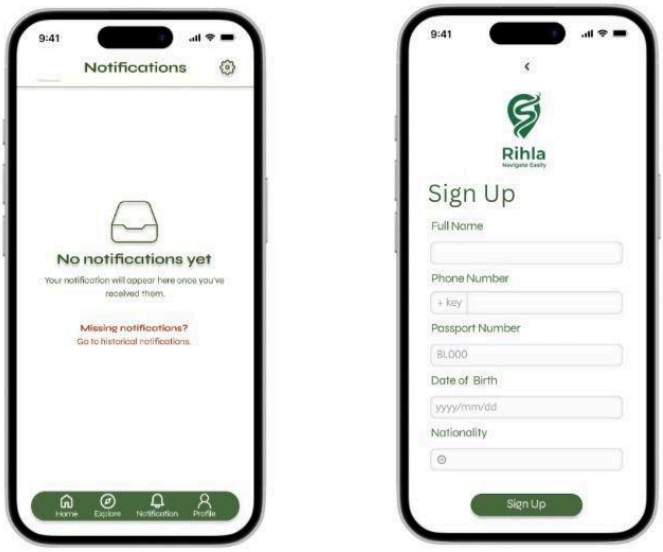






Rihla
Navigate Easily





Appendix C: To Be Determined List

- Third-party APIs to be decided during development.
- Transit providers and areas covered at the initial release.
- Pre-loaded tourist destinations for the first release.
- Offline capabilities and caching mechanism.